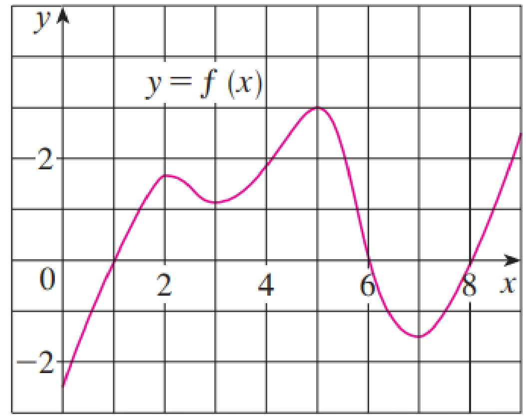
1. The graph of a function $f(x)$ is shown at the right. Assume that the domain of $f(x)$ is $[0,9]$. Use the graph of the function to help answer the following questions.



a) On what intervals is *f* increasing? Decreasing?

b) At what values of *x* does *f* have a local maximum? Local minimum?

c) Estimate on what intervals is *f* concave upward? Concave downward?

d) Estimate the *x*-coordinate(s) of the point(s) of inflection.

2. Consider the function $f(x)=x^4-\frac{8}{3}x^3+2x^2$. Without the use of a graph, determine each of the following.

a) On what intervals is *f* increasing? Decreasing?

b) At what values of *x* does *f* have a local maximum? Local minimum?

c) On what intervals is *f* concave upward? Concave downward?

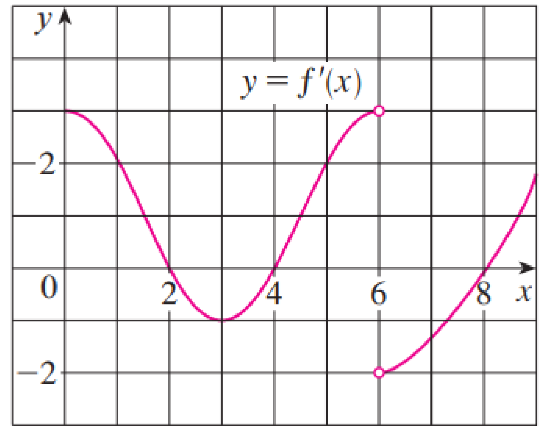
d) State the *x*-coordinate(s) of the point(s) of inflection.

3. Suppose the derivative of a function $f(x)$ is $f'(x)=\dfrac{(x+2)^2(x-3)^5(x-6)^4}{e^x\ln\left(x^2\right)}$.

a) Determine where $f(x)$ is increasing and decreasing, if you assume that $f(x)$ is defined on $(-\infty,\infty)$.

b) Determine at what *x*-values $f(x)$ has local extrema.

4. The graph of the derivative of $f'(x)$ is shown below.



a) On what intervals is f increasing? Decreasing?

b) At what values of x does f have a local maximum? Local minimum?

c) On what intervals is f concave upward? Concave downward?

d) State the x-coordinate(s) of the point(s) of inflection.

e) Use the information that you have to create a possible picture of $f(x)$.

5. Find the local maximum and minimum values of $f(x) = \sqrt{x} - \sqrt[4]{x}$ using both the First and Second Derivative Tests.

6. Suppose that $f(3) = 2$, $f'(3) = \frac{1}{2}$, and $f'(x) > 0$ and $f''(x) > 0$ for all x.

a) Sketch a possible graph for f.

b) How many solutions does the equation $f(x) = 0$ have? Why?

c) Is it possible that $f'(2) = \frac{1}{3}$?

7. Consider the function $f(x) = (x^2 + 1)e^{-x}$.

a) Determine the exact value of x at which $f(x)$ increases most rapidly. *Hint: Something is being maximized here. What function are we trying to maximize?*

b) Reflect on your work in part (a). What does your work here imply about the significance of inflection points?

8. Consider the function $f(x) = ax^3 + bx^2 + cx + d$, where *a, b, c,* & *d* are real numbers with $(a \neq 0)$.

a) Prove that $f(x)$ has exactly one inflection point.

b) Prove that if $f(x)$ has 3 *x*-intercepts, then the inflection point occurs at the average value of the intercepts.

c) Use the result to from part (b) to find the inflection point of the cubic polynomial $f(x) = x^3 - 3x^2 + 2x$, and check your result by finding the inflection points of $f(x)$ using the second derivative.

9. For what value *c* is the function $f(x) = cx + \dfrac{1}{x^2 + 3}$ increasing on $(-\infty, \infty)$.